

InvestSmart Test Cases

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Test Case 1: Open Home Page

Objective: Confirm that the system home page loads.

Precondition: Frontend files are available and Apache is running if testing locally.

Steps: Open <http://127.0.0.1/investsmart/public/index.html>.

Expected Result: Home page displays InvestSmart branding, Sign In button, Sign Up button, and system introduction.

Status: Pass if the page loads without broken layout.

Test Case 2: Client Sign Up

Objective: Confirm that a normal user can create an account.

Precondition: MySQL database exists and schema.sql has been executed.

Steps: Open Sign Up, enter full name, surname, ID number, email, password, and confirm password, then submit.

Expected Result: Account is created and user is routed or prompted to sign in.

Status: Pass if user record is created in users and clients tables.

Test Case 3: Duplicate Email Validation

Objective: Confirm duplicate accounts are prevented.

Precondition: A user already exists with the selected email.

Steps: Attempt to sign up using the same email.

Expected Result: System shows an error and does not create a duplicate user.

Status: Pass if only one user record exists for that email.

Test Case 4: Client Login with OTP

Objective: Confirm client login requires OTP after correct password.

Precondition: Client account exists.

Steps: Enter email and password, click Sign In, read the displayed OTP, enter OTP, click Verify OTP.

Expected Result: Client is redirected to dashboard.html.

Status: Pass if dashboard opens and client menu is visible.

Test Case 5: Admin Login from Same Login Page

Objective: Confirm admins use the same login page as clients.

Precondition: Default admin account exists.

Steps: Login using admin@investsmart.local and password, enter displayed OTP.

Expected Result: Admin is redirected to admin-dashboard.html.

Status: Pass if admin dashboard and admin menu are visible.

Test Case 6: Invalid Login

Objective: Confirm wrong password is rejected.

Precondition: Existing user account.

Steps: Enter correct email and wrong password.

Expected Result: System displays invalid email or password and does not show dashboard.

Status: Pass if no token/session is issued.

Test Case 7: Forgot Password with OTP

Objective: Confirm password reset flow works.

Precondition: Existing account.

Steps: Click Forgot Password, enter email, read displayed OTP, enter OTP and new password.

Expected Result: Password is updated and user can sign in with new password.

Status: Pass if old password no longer works and new password works.

Test Case 8: Profile View and Update

Objective: Confirm client profile details are displayed and editable.

Precondition: Client is logged in.

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Steps: Open Profile, update contact/email field, save.

Expected Result: Success message appears and database is updated.

Status: Pass if updated value persists after refresh.

Test Case 9: My Finances Net Salary Calculation

Objective: Confirm net salary is calculated in real time.

Precondition: Client is logged in.

Steps: Enter gross salary, deductions, monthly expenses, and savings.

Expected Result: Net salary updates dynamically and can be saved.

Status: Pass if financial_profiles contains saved values.

Test Case 10: Investment Calculator Recommendation

Objective: Confirm the calculator recommends a suitable bank.

Precondition: Bank seed data exists.

Steps: Enter plan name, amount, risk, horizon, liquidity, monthly contribution preference, and goal, then calculate.

Expected Result: System displays recommended bank with expected return, contact details, and investment link.

Status: Pass if recommendation is based on stored banks table data.

Test Case 11: Save Recommended Plan

Objective: Confirm a recommendation can be saved.

Precondition: Calculator has returned a recommendation.

Steps: Click Save Plan.

Expected Result: Plan is saved in investment_plans with bank_id and bank_investment_url.

Status: Pass if plan appears in My Plans.

Test Case 12: My Plans View, Edit, Delete

Objective: Confirm saved plans can be managed.

Precondition: Client has at least one saved plan.

Steps: Open My Plans, search/sort, view details, edit plan, delete plan.

Expected Result: Table updates and database reflects changes.

Status: Pass if actions work without affecting another user's plans.

Test Case 13: My Report

Objective: Confirm client report displays financial profile and saved plans.

Precondition: Client has finance data and saved plans.

Steps: Open My Report.

Expected Result: Report displays profile, financial overview, saved plans, provider data, and chart summaries.

Status: Pass if report can be printed/exported.

Test Case 14: Admin Dashboard Analytics

Objective: Confirm admin dashboard shows system metrics.

Precondition: Admin is logged in.

Steps: Open Admin Dashboard.

Expected Result: Total users, total plans, total savings, risk distribution, and analytics cards display.

Status: Pass if values match database data.

Test Case 15: Manage Users

Objective: Confirm admin can manage users.

Precondition: Admin is logged in and clients exist.

Steps: Open Manage Users, search user, edit user, activate/deactivate user, reset password.

Expected Result: Changes are saved and logged.

Status: Pass if activity_logs records admin actions.

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Test Case 16: Manage Providers/Banks

Objective: Confirm admin can manage bank/provider records.

Precondition: Admin is logged in.

Steps: Add a new bank, edit expected return/contact details, delete or deactivate provider if available.

Expected Result: Provider table updates and calculator can use the updated data.

Status: Pass if banks table changes are correct.

Test Case 17: Bank Link in Saved Plans

Objective: Confirm investment link is saved with the plan.

Precondition: Recommendation has a bank website.

Steps: Save a recommended plan, open My Plans, click or inspect provider link.

Expected Result: The investment URL is visible and points to the bank investment page.

Status: Pass if bank_investment_url is stored and shown.

Test Case 18: Role-Based Access Control

Objective: Confirm clients cannot access admin pages.

Precondition: Client is logged in.

Steps: Attempt to open admin-dashboard.html or admin API actions.

Expected Result: System blocks access or redirects to unauthorized page.

Status: Pass if client cannot view admin data.

Test Case 19: Audit Trail

Objective: Confirm important actions are logged.

Precondition: User and admin actions have been performed.

Steps: Check activity_logs table or Admin Activity view.

Expected Result: Logs show action, entity type, user/admin, description, and timestamp.

Status: Pass if logs are created for login, save, update, reset, and admin actions.

Test Case 20: Hosted Frontend with ngrok Backend

Objective: Confirm deployed frontend connects to local backend.

Precondition: XAMPP and ngrok are running and env.js contains ngrok API URL.

Steps: Open deployed frontend, sign in, complete OTP, open dashboard.

Expected Result: Frontend fetches backend data successfully.

Status: Pass if no Failed to fetch or invalid JSON errors appear.